

2015

Year 12 Mathematics Extension 2

Trial HSC Examination

Teacher Setting Paper: Mr P Mirrington
Head of Department: Mrs M Hill

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
(graphics calculators are not allowed)
- A table of standard integrals is provided at
the back of this paper
- For Section I record your answers on the
multiple choice answer sheet provided
- For Section II, answer each question in a
SEPARATE writing booklet. Extra writing
booklets are available.
- For Section II show relevant mathematical
reasoning and/or calculations

Total Marks – 100

Section I Pages 2 – 5

10 Marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II Pages 6 – 13

90 Marks

- Attempt Questions 11 – 16
- Each Question is worth 15 marks
- Allow about 2 hours and 45 minutes for
this section

*This examination paper does not necessarily reflect the content or
format of the Higher School Certificate Examination in this subject*

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet to record your answers for Questions 1–10.

QUESTION 1

The eccentricity of the hyperbola $x^2 - 4y^2 = 1$ is:

- (A) $\frac{4}{5}$
- (B) $\frac{2}{\sqrt{5}}$
- (C) $\frac{\sqrt{5}}{2}$
- (D) $\frac{5}{4}$

QUESTION 2

The *acute* angle between the asymptotes of the hyperbola $x^2 - \frac{y^2}{3} = 1$ is:

- (A) $\frac{\pi}{3}$
- (B) $\frac{2\pi}{3}$
- (C) $\frac{\pi}{6}$
- (D) $\frac{\pi}{4}$

QUESTION 3

P, Q and R are three consecutive terms in an arithmetic progression.

Which of the following is the simplification of $\frac{\sin(P+R)}{\cos Q}$?

- (A) $2\sin Q$
- (B) $2\cos Q$
- (C) $\sin 2Q$
- (D) $\cos 2Q$

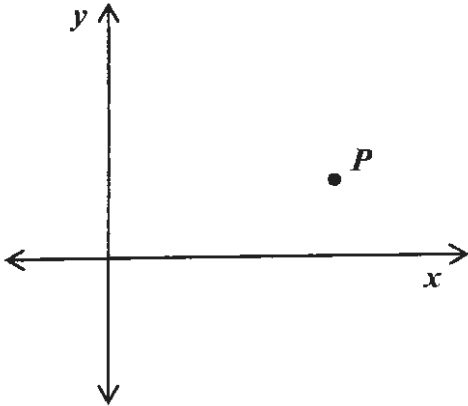
QUESTION 4

How many asymptotes does the graph of the curve $y = \frac{x-1}{x^2-1}$ have?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

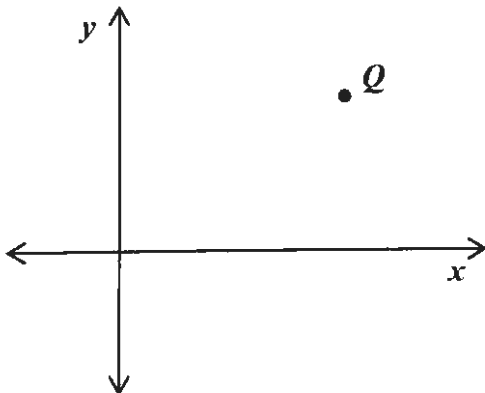
QUESTION 5

On the Argand diagram below, P represents the complex number z .

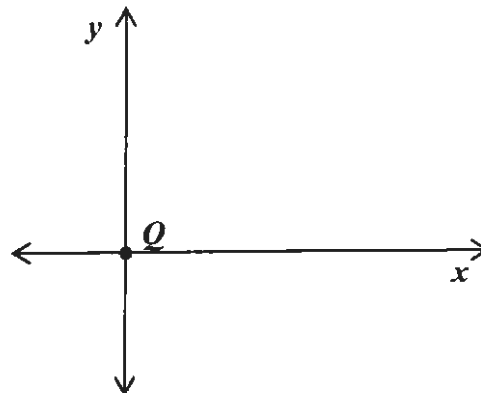


Which of the following Argand diagrams shows the point Q representing $z - \bar{z}$?

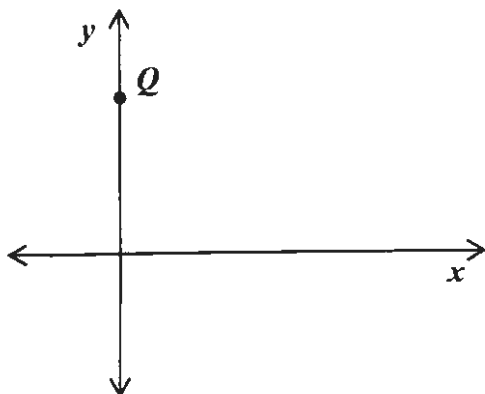
(A)



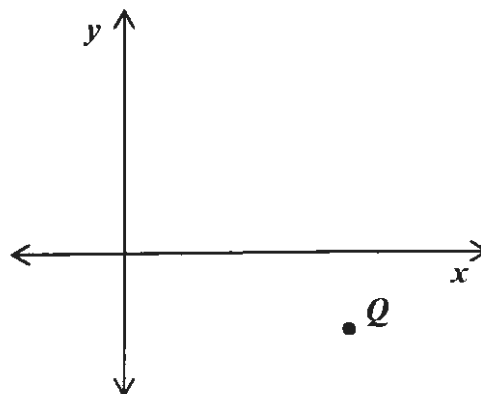
(B)



(C)



(D)



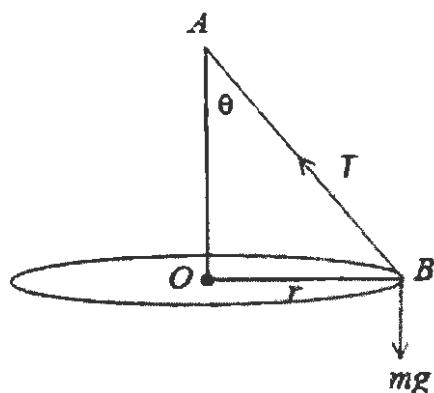
QUESTION 6

The equation $x^3 + 3x - 2 = 0$ has roots α, β and γ . The value of $\alpha^3 + \beta^3 + \gamma^3$ is:

- (A) 2
- (B) 6
- (C) -2
- (D) -6

QUESTION 7

A particle of mass m at B is attached to a string AB that is fixed at A . The particle rotates in a horizontal circle with a radius of r . Let T be the tension in the string and $\angle BAO = \theta$.



Which of the following statements is *not* correct:

- (A) $T > mg$
- (B) $T = \frac{m\omega^2 r}{\sin \theta}$
- (C) $r = \frac{g \tan \theta}{\omega^2}$
- (D) $\omega = \sqrt{\frac{g \cot \theta}{r}}$

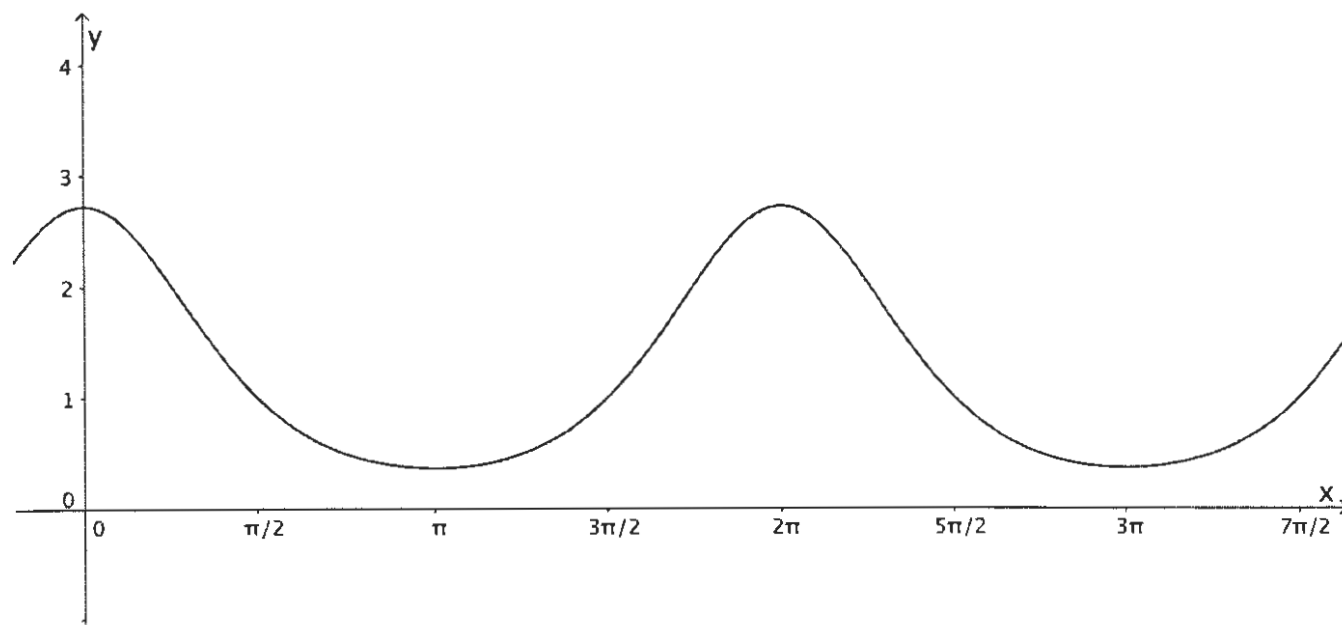
QUESTION 8

Consider the rectangular hyperbola $xy = 16$. If this hyperbola was rotated 45 degrees in a clockwise direction, its equation would be:

- (A) $x^2 - y^2 = 8$
- (B) $x^2 - y^2 = 16$
- (C) $x^2 - y^2 = 16\sqrt{2}$
- (D) $x^2 - y^2 = 32$

QUESTION 9

Which function best represents the graph below?



- (A) $y = e \cos x$
- (B) $y = e + 1 - \cos x$
- (C) $y = \frac{e}{\cos x}$
- (D) $y = e^{\cos x}$

QUESTION 10

When $y^2 + xy = 1$ is differentiated with respect to x , the correct expression for the derivative is:

- (A) $\frac{dy}{dx} = \frac{1}{1+2y}$
- (B) $\frac{dy}{dx} = \frac{1-y}{1+2y}$
- (C) $\frac{dy}{dx} = \frac{-y}{x+2y}$
- (D) $\frac{dy}{dx} = \frac{1-y}{x+2y}$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a separate writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

QUESTION 11 (15 Marks) Use a SEPARATE writing booklet.

Marks

(a) Let $z = 1 - 3i$ and $\omega = 2 + i$.

Find the value of $\frac{z}{\omega}$ in the form $a + ib$, where a and b are real numbers:

2

(b) (i) If $z = x + iy$, write down expressions for $z\bar{z}$ and $z + \bar{z}$.

1

(ii) Sketch the region on the Argand diagram described by: $z\bar{z} \leq 3(z + \bar{z})$

2

(c) Factorise $P(x) = x^3 + x^2 + 4x + 4$ into its three linear factors.

2

(d) Find these indefinite integrals:

(i) $\int \frac{1}{x^2 + 2x + 5} dx$

2

(ii) $\int \frac{1}{x^2 + 2x - 3} dx$

2

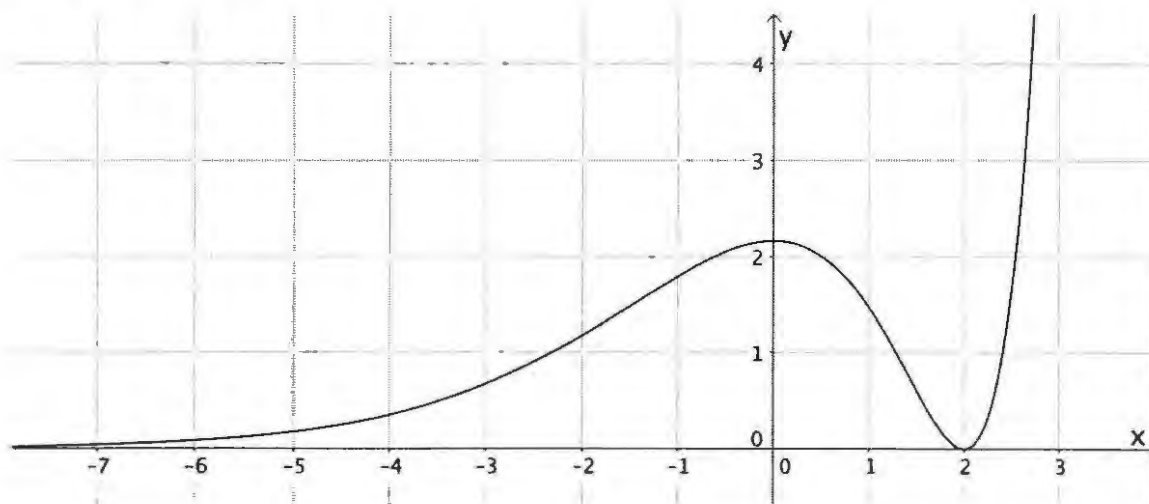
(e) (i) Find all pairs of integers a and b such that $(a + ib)^2 = 7 + 24i$

2

(ii) Hence, or otherwise, solve: $z^2 + 3iz - (4 + 6i) = 0$

2

- (a) The function $f(x) = 4(x-2)^2 e^{x-2}$ has stationary points at $x = 0$ and $x = 2$ as shown in the diagram below.



Draw separate half-page sketches of the following graphs. In each case, label any asymptote and the coordinates of any turning points.

(i) $y = \frac{1}{f(x)}$

2

(ii) $y = \sqrt{f(x)}$

2

(iii) $y = f(|x|)$

1

- (b) (i) Use the result $\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$ to show that

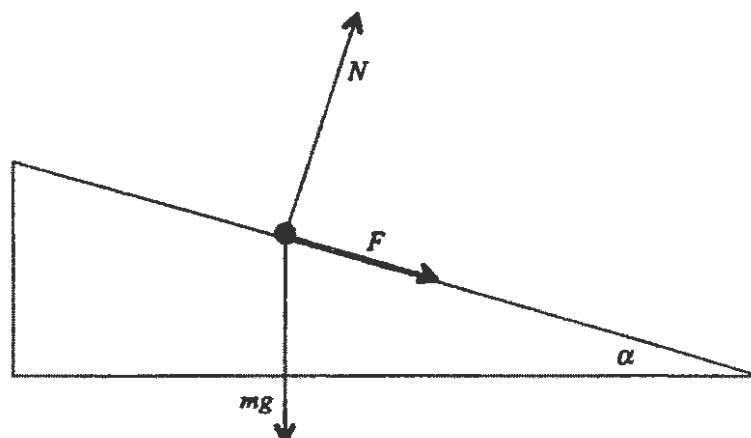
$$\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right) \quad 2$$

- (ii) Hence find the general solution to the equation: $\sin 2x + \sin 4x = \sin 6x$ 3

- (iii) Use part (i) to evaluate $\int_0^{\frac{\pi}{4}} \sin 5x \cos 3x \, dx$ 3

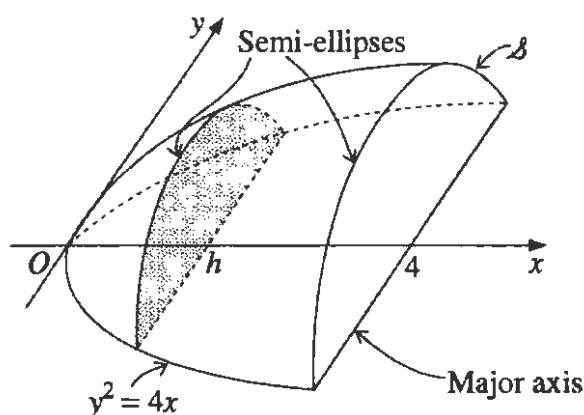
- (c) If $A(x)$ and $B(x)$ are *odd* polynomial functions show that the product $P(x) = A(x) \times B(x)$ is an *even* polynomial function. 2

- (a) A car of mass 800kg moves with constant speed in a horizontal circle of radius 250m on a track that is banked at an angle of $\alpha = 7^\circ$ to the horizontal. The forces acting on the car are the weight force mg , a normal reaction force N to the road and a friction force F acting down the slope as shown. The acceleration due to gravity is 9.8 ms^{-2} .



- (i) Show that the optimum speed for the car (when the friction force F acting on the car is zero) is about 62 km/hr . 2
- (ii) If the car is travelling at a constant speed of 108 km/hr , find the size of the friction force F acting on the car? (Give your answer to the nearest Newton) 3

(b)

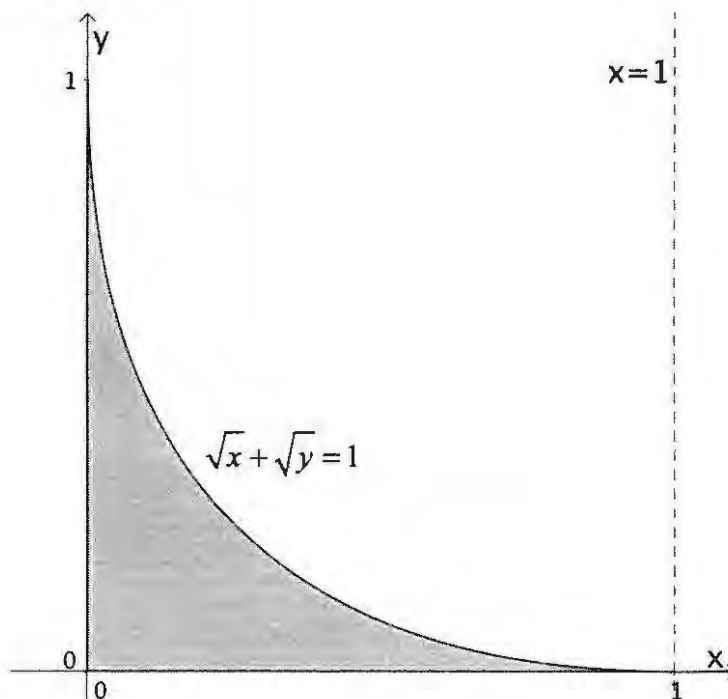


The base of a solid S is the region in the xy plane enclosed by the parabola $y^2 = 4x$ and the line $x = 4$, and each cross section perpendicular to the x axis is a semi-ellipse with the minor axis one-half of the major axis.

- (i) Show that the area of the semi-ellipse at $x = h$ is πh .
(You may assume that the area of an ellipse with semi-axes a and b is πab .) 2
- (ii) Find the volume of the solid S . 2

QUESTION 13 is continued on the next page.

- (c) The region bounded by the curve $\sqrt{x} + \sqrt{y} = 1$ and the x axis between $x = 0$ and $x = 1$ is rotated about the line $x = 1$.



- (i) Use the method of cylindrical shells to show that the volume V of the solid formed is given by:

$$V = 2\pi \int_0^1 1 - x^2 + 2x\sqrt{x} - 2\sqrt{x} \, dx \quad 2$$

- (ii) Hence find the value of V in simplest form 1

- (d) The equation $x^3 + 2x^2 + 3x + 4 = 0$ has roots α, β and γ .

Find the monic cubic equation with roots α^2, β^2 and γ^2 . 3

(a) The complex number ω is given by $\omega = 2(\cos \theta + i \sin \theta)$, where $\frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}$.

(i) Express each of ω^2 and $\frac{1}{\omega}$ in modulus/argument form. 2

(ii) On an Argand diagram, the points Q and R represent the complex numbers ω^2 and $\frac{1}{\omega}$ respectively. If the points Q, O and R are collinear, find ω in the form $a + ib$, where a and b are real numbers. 2

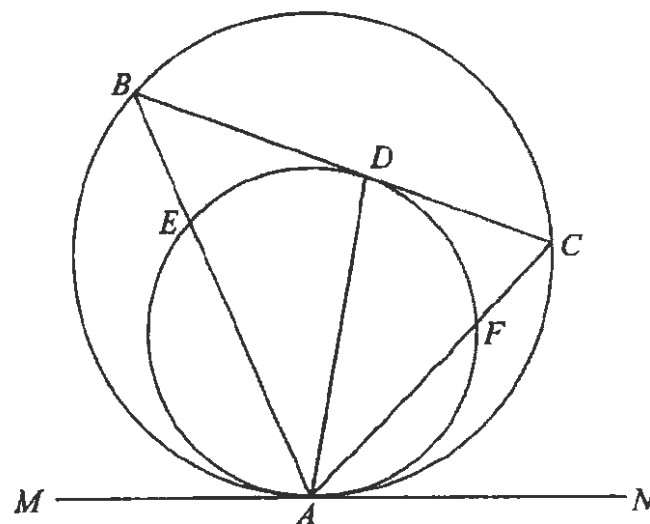
(b) Let $I_n = \int_0^a (1+x^2)^n dx$, $n = 1, 2, 3, \dots$

(i) Use integration by parts to show that $I_n = \frac{a(1+a^2)^n}{2n+1} + \frac{2n}{2n+1} I_{n-1}$ 3

(ii) Show that $I_1 = \frac{3a+a^3}{3}$ 2

(iii) Hence, or otherwise, evaluate $\int_0^1 (1+x^2)^3 dx$ 2

(c)



In the diagram, MAN is the common tangent to two circles touching internally at A. B and C are two points on the larger circle such that BC is a tangent to the smaller circle with point of contact D. AB and AC cut the smaller circle at E and F respectively.

Copy the diagram. Show that AD bisects $\angle BAC$.

4

- (a) A particle of mass m kg is dropped from rest in a medium where the resistance to the motion has magnitude $\frac{1}{40}mv^2$ when the speed of the particle is v ms⁻¹. After t seconds the particle has fallen x metres. The acceleration due to gravity is 10 ms⁻².

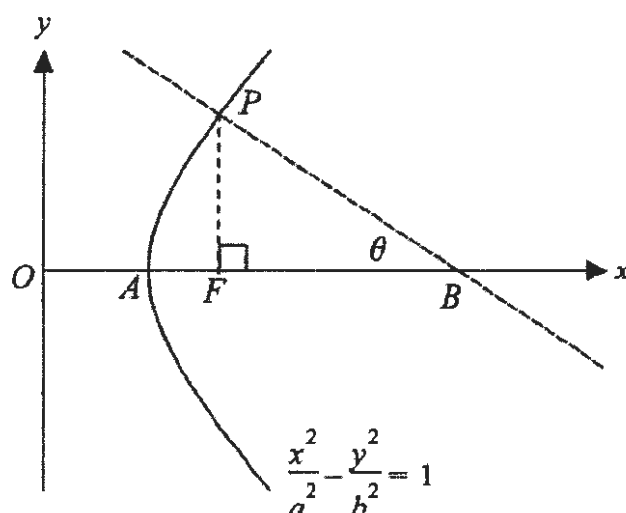
(i) Explain why $\ddot{x} = \frac{1}{40}(400 - v^2)$. 1

(ii) Find an expression for t in terms of v by integration. 2

(iii) Show that $v = 20\left(1 - \frac{2}{1 + e^t}\right)$. 1

- (b) Consider the polynomial $P(x) = x^4 - 2x^3 - x^2 + 6x - 6$ over the complex field. Given that $P(1 - i) = 0$, find all four solutions to $P(x) = 0$. 2

- (c) In the diagram below, F is a focus of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with eccentricity e .



This branch of the hyperbola cuts the x axis at A where $AF = h$. P is the point on the hyperbola vertically above F and the normal at P cuts the x axis at B making an acute angle θ with the x axis.

(i) Show that $\tan \theta = \frac{1}{e}$ 3

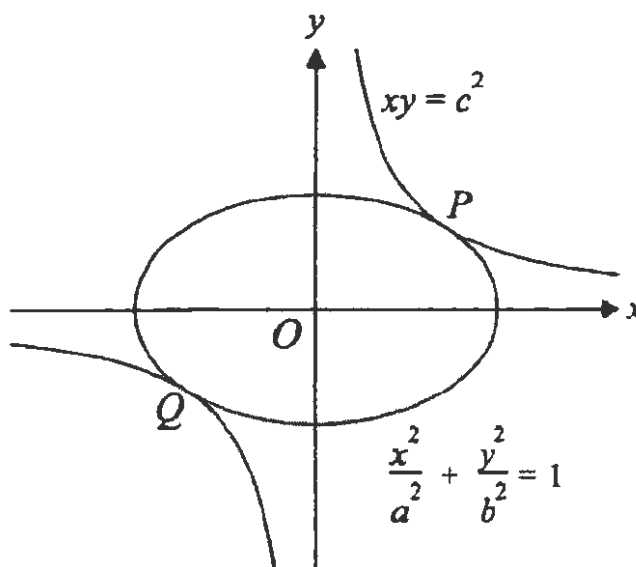
(ii) Show that $PF = h(e + 1)$ 1

A bowl is formed by rotating the hyperbola above through one revolution about the x axis. The bowl is then placed on a horizontal table with point A on the table. A particle P of mass m is set in motion around the inside of the bowl, travelling with constant angular velocity ω in a horizontal circle with centre F .

(iii) Show that $\omega^2 = \frac{g}{he(e + 1)}$ 3

(iv) N is the normal reaction force between the particle P and the bowl. Show that if the hyperbola used to form the bowl is a rectangular hyperbola, then $N = mg\sqrt{\frac{3}{2}}$. 2

- (a) The hyperbola $xy = c^2, c > 0$ touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b > 0$ at points P and Q, where $P\left(cp, \frac{c}{p}\right)$ lies in the first quadrant.



$\left(ct, \frac{c}{t}\right)$ is the general parametrisation of $xy = c^2, t \neq 0$

- (i) Explain why the equation $(bc)^2 t^4 - (ab)^2 t^2 + (ca)^2 = 0$ has roots $p, p, -p, -p$ where $p > 0$. 2
- (ii) Deduce that $p = \frac{a}{c\sqrt{2}}$ and $ab = 2c^2$. 2
- (iii) Show that if $S(c\sqrt{2}, c\sqrt{2})$ and S' are the foci of the hyperbola $xy = c^2$, then the quadrilateral with vertices P, S, Q and S' has area $2c(a - b)$. 2
- (b) If $a > 0$ and $b > 0$, prove that:
- (i) $\frac{1}{a} + \frac{1}{b} \geq \frac{4}{a+b}$. 2
- (ii) $\frac{1}{a^2} + \frac{1}{b^2} \geq \frac{8}{(a+b)^2}$. 2

QUESTION 16 is continued on the next page.

(c) Let $f(x) = \frac{x^n}{e^x}$, where $n > 1$.

(i) Show that $f'(x) = \frac{x^{n-1}(n-x)}{e^x}$ 1

(ii) Show that the graph of $y = f(x)$ has a maximum turning point at $\left(n, \frac{n^n}{e^n}\right)$, and hence sketch the graph for $x \geq 0$. (Don't attempt to find any points of inflexion). 2

(iii) Explain, by considering the graph of $y = f(x)$ for $x > n$, why $\frac{x^n}{e^x} < \frac{n^n}{e^n}$ for $x > n$. 1

(iv) Deduce from part (iii), using the substitution $x = n + 1$, that $\left(1 + \frac{1}{n}\right)^n < e$. 1

End of examination

$$1. \quad \frac{x^2}{12} - \frac{y^2}{(y_2)^2} = 1 \quad a=1 \quad b=\frac{1}{2}$$

$$b^2 = a^2(e^2 - 1)$$

$$\frac{1}{4} = 1(e^2 - 1)$$

$$e^2 = \frac{5}{4}$$

$$e = \frac{\sqrt{5}}{2}$$

C

$$2. \quad \frac{x^2}{1} - \frac{y^2}{3} = 1 \quad a=1 \quad b=\sqrt{3}$$

asymptotes $y = \pm \frac{b}{a} x$
 $y = \pm \sqrt{3} x$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\sqrt{3} - (-\sqrt{3})}{1 - 3} \right|$$

$$= \sqrt{3}$$

$$\theta = \pi/3$$

A

$$3. \quad P, Q, R \text{ in AP}$$

$$\therefore Q = \frac{P+R}{2}$$

$$P+R = 2Q$$

$$\frac{\sin(P+R)}{\cos Q} = \frac{\sin 2Q}{\cos Q}$$

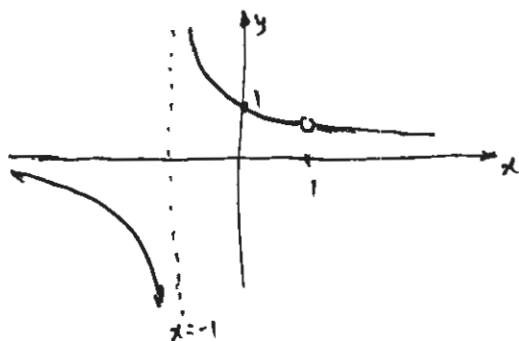
$$= \frac{2 \sin Q \cos Q}{\cos Q}$$

$$= 2 \sin Q$$

A

$$4. \quad y = \frac{x-1}{x^2-1} = \frac{x-1}{(x+1)(x-1)}$$

$$= \frac{1}{x+1}$$



2 asymptotes.

B

$$5. \quad z - \bar{z} = (x+iy) - (x-iy) \\ = 2iy$$

$\therefore Q$ has x coord = zero
 y coord = $2 \times y$ coord of P .

C

$$6. \quad x^3 + 3x - 2 = 0$$

$$\alpha^3 + 3\alpha - 2 = 0$$

$$\beta^3 + 3\beta - 2 = 0$$

$$\gamma^3 + 3\gamma - 2 = 0$$

$$\left. \begin{array}{l} \alpha^3 + 3\alpha - 2 = 0 \\ \beta^3 + 3\beta - 2 = 0 \\ \gamma^3 + 3\gamma - 2 = 0 \end{array} \right\} \text{ add } \Rightarrow \alpha^3 + \beta^3 + \gamma^3 + 3(\alpha + \beta + \gamma) - 6 = 0$$

$$\alpha^3 + \beta^3 + \gamma^3 + 3 \times 0 = 6$$

$$\alpha^3 + \beta^3 + \gamma^3 = 6$$

B

$$7. \quad T \cos \theta = mg$$

$$T = \frac{mg}{\cos \theta}$$

$$\therefore T > mg \quad \text{since } \cos \theta < 1$$

$$T \sin \theta = m\omega^2 r \Rightarrow T = \frac{m\omega^2 r}{\sin \theta} \checkmark$$

$$\therefore \tan \theta = \frac{\omega^2 r}{g}$$

$$r = \frac{g \tan \theta}{\omega^2} \checkmark$$

$$\omega^2 = \frac{g \tan \theta}{r}$$

$\therefore D$ is not correct.

D

$$8. \quad xy = 16 = c^2 = \frac{a^2}{2}$$

$$\therefore a^2 = 32$$

$$x^2 - y^2 = 32$$

D

9. $\cos 0 = 1$
 $\cos \pi = -1$ eliminates A and B
 $\cos \frac{\pi}{2} = 0$ eliminates C

D

10. $y^2 + xy = 1$

$$2y \frac{dy}{dx} + x \frac{dy}{dx} + y = 0$$

$$\frac{dy}{dx} (2y + x) + y = 0$$

$$\frac{dy}{dx} = \frac{-y}{2y+x}$$

C

QUESTION 11

(a) $z = 1 - 3i$
 $w = 2 + i$

$$\frac{z}{w} = \frac{1 - 3i}{2 + i} \times \frac{2 - i}{2 - i}$$

$$= \frac{2 - i - 6i + 3i^2}{2^2 - i^2}$$

$$= \frac{2 - 7i - 3}{4 - (-1)}$$

$$= \frac{-1 - 7i}{5}$$

$$= -\frac{1}{5} - \frac{7i}{5}$$

(b) (i) $z = x + iy$

$$z\bar{z} = (x + iy)(x - iy)$$

$$= x^2 - i^2 y^2$$

$$= x^2 + y^2$$

$$z + \bar{z} = x + iy + x - iy$$

$$= 2x$$

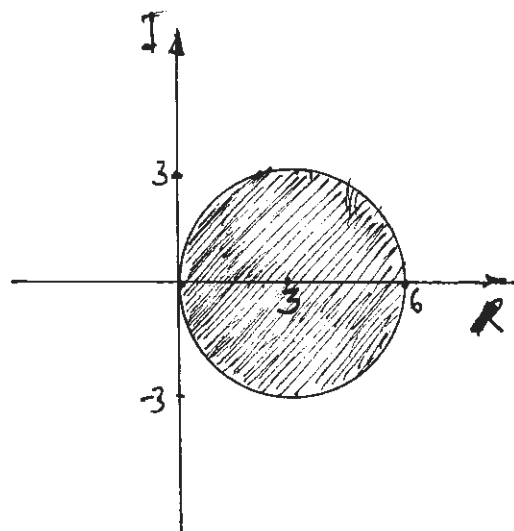
(ii) $z\bar{z} \leq 3(z + \bar{z})$

$$x^2 + y^2 \leq 3(2x)$$

$$x^2 - 6x + y^2 \leq 0$$

$$(x - 3)^2 + y^2 \leq 9$$

circle centre (3, 0) radius 3.



(c) $P(x) = x^3 + x^2 + 4x + 4$

$$P(-1) = -1 + 1 - 4 + 4 = 0$$

$\therefore (x + 1)$ is a factor.

$$P(x) = (x + 1)(x^2 + 4) \text{ by inspection (or division)}$$

$$= (x + 1)(x - 2i)(x + 2i)$$

QUESTION 11 (continued)

$$(d) (i) \int \frac{1}{x^2 + 2x + 5} dx = \int \frac{1}{(x+1)^2 + 4} dx$$

$$= \int \frac{1}{u^2 + 2^2} du \quad \begin{array}{l} u = x+1 \\ \frac{du}{dx} = 1 \\ du = dx \end{array}$$

$$= \frac{1}{2} \tan^{-1}\left(\frac{u}{2}\right) + C$$

$$= \frac{1}{2} \tan^{-1}\left(\frac{x+1}{2}\right) + C$$

$$(ii) \int \frac{1}{x^2 + 2x - 3} dx = \int \frac{1}{(x+3)(x-1)} dx$$

$$\text{Let } \frac{1}{(x+3)(x-1)} = \frac{a}{(x+3)} + \frac{b}{(x-1)}$$

$$1 = a(x-1) + b(x+3)$$

$$x=1 \Rightarrow 1 = 4b$$

$$b = \frac{1}{4}$$

$$x=-3 \Rightarrow 1 = -4a$$

$$a = -\frac{1}{4}$$

$$\left. \begin{array}{l} 1 = 4b \\ b = \frac{1}{4} \\ 1 = -4a \\ a = -\frac{1}{4} \end{array} \right\} \therefore \frac{1}{(x+3)(x-1)} = \frac{1}{4} \left(\frac{1}{x-1} - \frac{1}{x+3} \right)$$

$$\therefore \int \frac{1}{x^2 + 2x - 3} dx = \frac{1}{4} \int \left(\frac{1}{x-1} - \frac{1}{x+3} \right) dx$$

$$= \frac{1}{4} \ln(x-1) - \frac{1}{4} \ln(x+3) +$$

$$= \frac{1}{4} \ln\left(\frac{x-1}{x+3}\right) + C$$

QUESTION 11 (continued)

$$(e) \quad (i) \quad (a+ib)^2 = 7+24i$$
$$a^2 - b^2 + 2abi = 7 + 24i$$

equating real & imaginary parts:

$$a^2 - b^2 = 7 \quad \text{and} \quad 2ab = 24$$

$$ab = 12$$

$$a = \frac{12}{b}$$

$$\left(\frac{12}{b}\right)^2 - b^2 = 7 \quad \xleftarrow{\text{substitute}}$$

$$144 - b^4 = 7b^2$$

$$b^4 + 7b^2 - 144 = 0$$

$$(b^2 + 16)(b^2 - 9) = 0$$

$$b = \pm 3 \quad a = \pm 4$$

Solutions: $a=4 \quad b=3$ OR $a=-4 \quad b=-3$

$$(ii) \quad z^2 + 3iz - (4+6i) = 0$$

$$z = \frac{-3i \pm \sqrt{(3i)^2 - 4(4+6i)}}{2}$$

$$= \frac{-3i \pm \sqrt{-9 + 16 + 24i}}{2}$$

$$= \frac{-3i \pm \sqrt{7+24i}}{2}$$

$$= \frac{-3i \pm (4+3i)}{2}$$

from part (i)

$$= \frac{-3i + 4 + 3i}{2}$$

OR

$$\frac{-3i - 4 - 3i}{2}$$

$$= 2$$

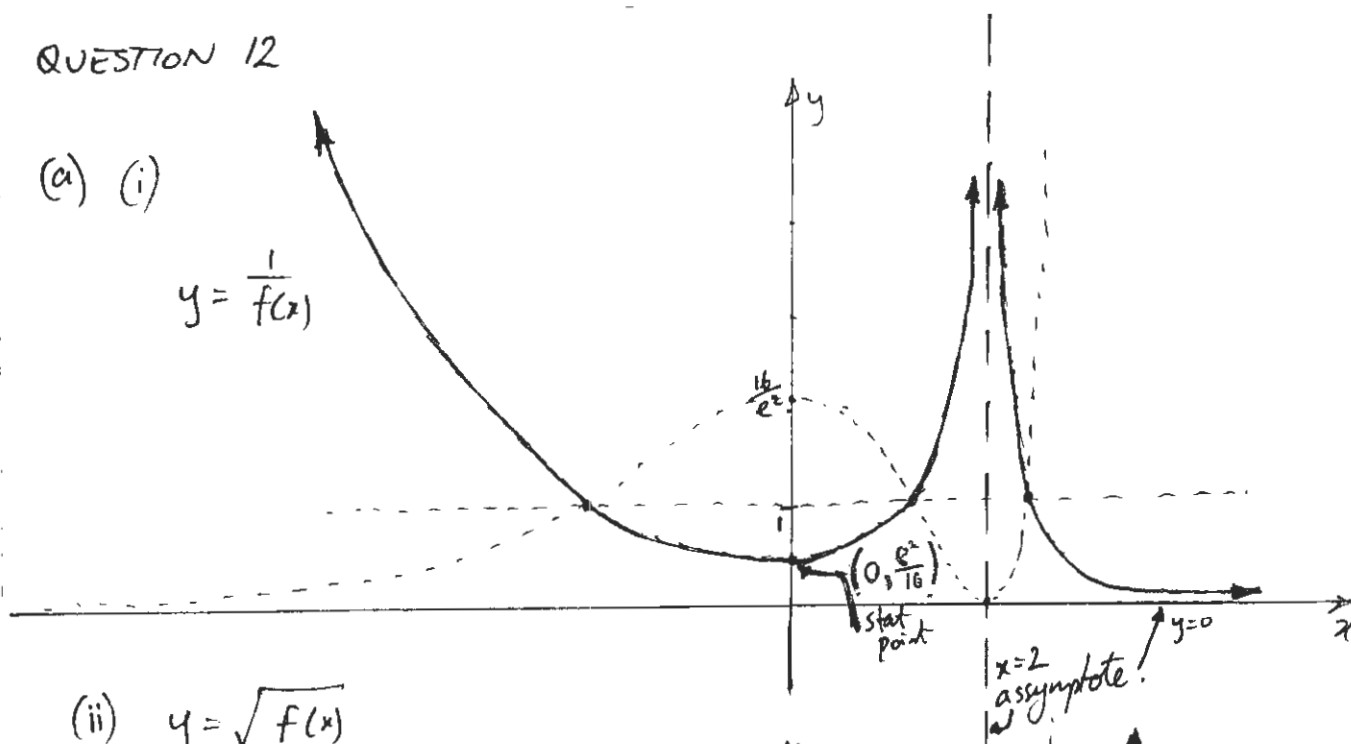
OR

$$-2 - 3i$$

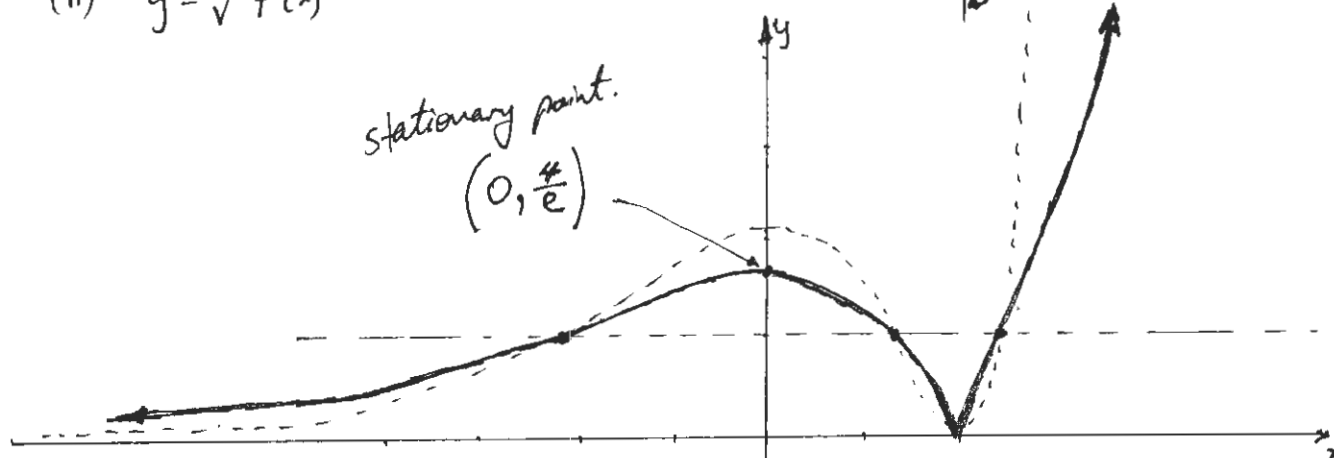
QUESTION 12

(a) (i)

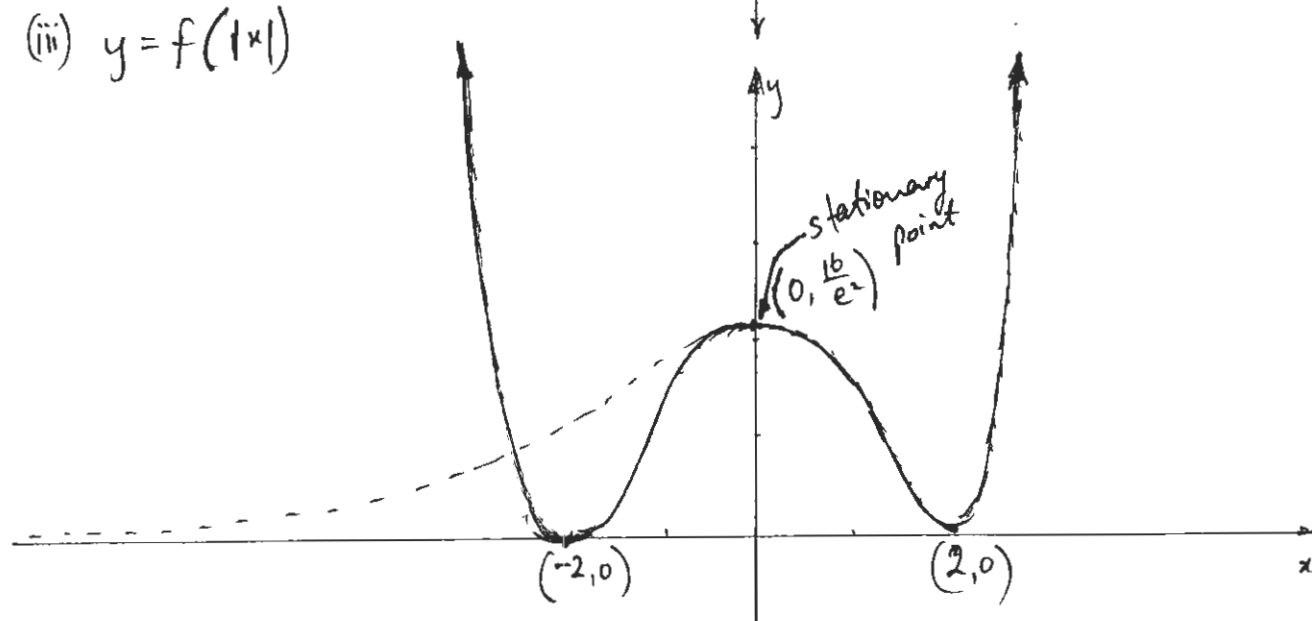
$$y = \frac{1}{f(x)}$$



(ii) $y = \sqrt{f(x)}$



(iii) $y = f(|x|)$



QUESTION 12 (continued)

(b) (i) $\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta.$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta \quad \text{--- (1)}$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta. \quad \text{--- (2)}$$

$$\textcircled{1} + \textcircled{2} \quad \sin(\alpha + \beta) + \sin(\alpha - \beta) = 2 \sin \alpha \cos \beta \quad \text{--- (3)}$$

Let $A = \alpha + \beta$

$B = \alpha - \beta$

$\therefore 2\alpha = A + B$

$2\beta = A - B$

$\alpha = \frac{A+B}{2}$

$\beta = \frac{A-B}{2}$

substitute into (3) $\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right).$

(ii) $\sin 2x + \sin 4x = \sin 6x$
 $2 \sin\left(\frac{2x+4x}{2}\right) \cos\left(\frac{2x-4x}{2}\right) = \sin 6x$ using (i) with $A=2x$
 $2 \sin 3x \cos(-x) = \sin 6x$ $B=4x$
 $2 \sin 3x \cos x = 2 \sin 3x \cos 3x$
 $2 \sin 3x (\cos x - \cos 3x) = 0$
 $\sin 3x = 0$ or $\cos 3x = \cos x$
 $3x = \pi n$ or $3x = 2\pi n \pm x$
 $x = \frac{\pi n}{3}$ or $2x = 2\pi n$ or $4x = 2\pi n$ $n = \pm 1, \pm 2, \pm 3, \dots$
 $x = \frac{\pi n}{3}$ or $x = \frac{\pi n}{2}$

(iii) $\int_0^{\pi/4} \sin 5x \cos 3x \, dx = \frac{1}{2} \int_0^{\pi/4} 2 \sin\left(\frac{8x+2x}{2}\right) \cos\left(\frac{8x-2x}{2}\right) \, dx$ $\frac{A+B}{2} = 5x$
 $= \frac{1}{2} \int_0^{\pi/4} \sin 8x + \sin 2x \, dx$ $A+B = 10x$
 $= \frac{1}{2} \left[-\frac{1}{8} \cos 8x + \frac{1}{2} \cos 2x \right]_0^{\pi/4}$ $\frac{A-B}{2} = 3x$
 $= -\frac{1}{2} \left(\frac{1}{8} + 0 - \left(\frac{1}{8} + \frac{1}{2} \right) \right) = \frac{1}{4}$ $A-B = 6x$
 $A = 8x, B = 2x$

QUESTION 12 (continued)

$$(c) \quad \begin{array}{l} A(-x) = -A(x) \\ B(-x) = -B(x) \end{array} \quad \left. \vphantom{\begin{array}{l} A(-x) = -A(x) \\ B(-x) = -B(x) \end{array}} \right\} \text{ since odd functions.}$$

$$P(x) = A(x) \times B(x)$$

$$P(-x) = A(-x) \times B(-x)$$

$$= -A(x) \times -B(x)$$

$$= A(x) \times B(x)$$

$$= P(x)$$

$\therefore P(x)$ is even.

QUESTION 13

(a) (i) Vertically

$$N \cos \alpha - F \sin \alpha - mg = 0$$

$$N \cos \alpha = F \sin \alpha + mg \quad \text{--- (1)}$$

Radially

$$F \cos \alpha + N \sin \alpha = \frac{mv^2}{r}$$

$$N \sin \alpha = \frac{mv^2}{r} - F \cos \alpha \quad \text{--- (2)}$$

$$\text{If } F = 0 \quad N \cos \alpha = mg \quad \text{--- (3)}$$

$$N \sin \alpha = \frac{mv^2}{r} \quad \text{--- (4)}$$

$$\text{(4)} \div \text{(3)} \quad \tan \alpha = \frac{v^2}{gr}$$

$$\therefore v^2 = gr \tan \alpha$$

$$= 9.8 \times 250 \times \tan 7^\circ$$

$$v = \sqrt{9.8 \times 250 \times \tan 7^\circ} \text{ m/s}$$

$$\approx \sqrt{300.8} \text{ m/s}$$

$$\approx 17.344 \text{ m/s}$$

$$= 17.344 \times 3.6 \text{ km/h}$$

$$\approx 62.4 \text{ km/h as required.}$$

(ii) (2) $\div$ (1) gives

$$\tan \alpha = \frac{\frac{mv^2}{r} - F \cos \alpha}{F \sin \alpha + mg}$$

$$F \frac{\sin^2 \alpha}{\cos \alpha} + mg \frac{\sin \alpha}{\cos \alpha} = \frac{mv^2}{r} - F \cos \alpha$$

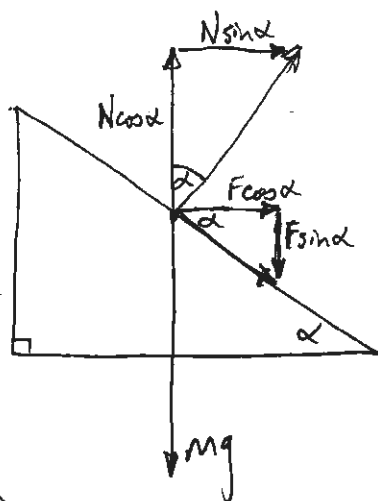
$$F \sin^2 \alpha + mg \sin \alpha = \frac{mv^2}{r} \cos \alpha - F \cos^2 \alpha$$

$$F(\sin^2 \alpha + \cos^2 \alpha) = \frac{mv^2 \cos \alpha}{r} - mg \sin \alpha$$

$$F = \frac{mv^2 \cos \alpha}{r} - mg \sin \alpha$$

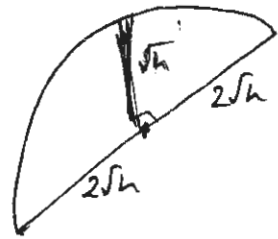
$$= \frac{800 \times 30^2 \cos 7^\circ}{250} - 800 \times 9.8 \times \sin 7^\circ$$

$$= 2858.53... - 955.45... = 1903 \text{ Newtons (nearest newton)}$$



QUESTION 13 (continued)

(b) (i) when $x=h$ $y^2 = 4h$
 $y = \pm 2\sqrt{h}$



$\therefore$ the base of the semi-ellipse is $4\sqrt{h}$

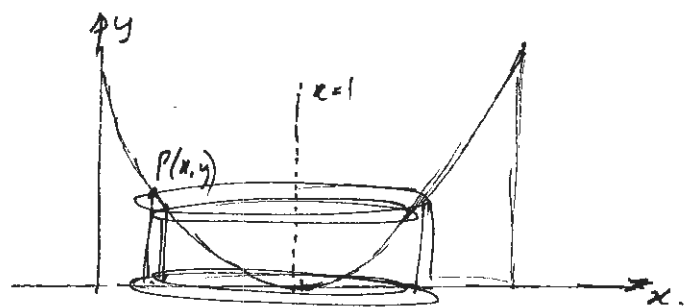
$\therefore$ The area of the semi-ellipse $= \frac{1}{2} \pi ab$
 $= \frac{1}{2} \pi (2\sqrt{h})(\sqrt{h})$
 $= \frac{1}{2} \times 2\pi (\sqrt{h})^2$
 $= \pi h$ (as required).

(ii) Volume of solid $= \sum_{h=0}^4 \text{Area} \times \delta h$
 $= \int_0^4 \pi h \, dh$
 $= \pi \left[\frac{h^2}{2} \right]_0^4$
 $= \pi \left(\frac{16}{2} - 0 \right)$
 $= 8\pi \text{ units}^3.$

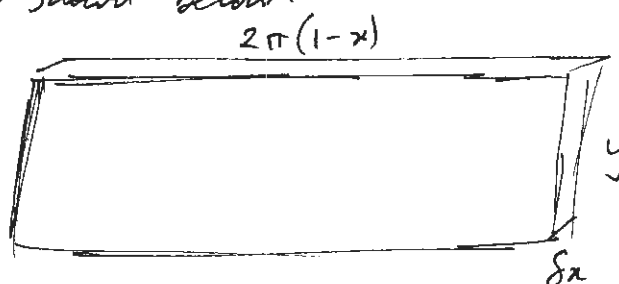
(c) over page

QUESTION 13 (continued)

(c) (i)



The cylindrical shell opens out to form a thin rectangular prism as shown below:



$$\begin{aligned} \delta V &= 2\pi(1-x)y \delta x \\ &= 2\pi(1-x)(1-\sqrt{x})^2 \delta x \end{aligned}$$

$$\begin{aligned} \sqrt{y} &= 1-\sqrt{x} \\ y &= (1-\sqrt{x})^2 \end{aligned}$$

$$V = 2\pi \int_0^1 (1-x)(1-\sqrt{x})^2 dx$$

$$= 2\pi \int_0^1 (1-x)(1-2\sqrt{x}+x) dx$$

$$= 2\pi \int_0^1 1-2\sqrt{x}+x-x+2x\sqrt{x}-x^2 dx$$

$$= 2\pi \int_0^1 1-x^2+2x\sqrt{x}-2\sqrt{x} dx \quad \text{as required.}$$

$$(ii) \quad V = 2\pi \left[x - \frac{x^3}{3} + \frac{2x^{5/2}}{5/2} - \frac{2x^{3/2}}{3/2} \right]_0^1$$

$$= 2\pi \left[x - \frac{x^3}{3} + \frac{4x^{5/2}}{5} - \frac{4x^{3/2}}{3} \right]_0^1$$

$$= 2\pi \left(1 - \frac{1}{3} + \frac{4}{5} - \frac{4}{3} \right)$$

$$= \frac{4\pi}{15} \text{ units}^3.$$

QUESTION 13 (continued)

(d) $P(x) = x^3 + 2x^2 + 3x + 4 = 0$ roots α, β, γ

$P(\sqrt{x}) = 0$ will have roots $\alpha^2, \beta^2, \gamma^2$

$$(\sqrt{x})^3 + 2(\sqrt{x})^2 + 3(\sqrt{x}) + 4 = 0$$

$$x\sqrt{x} + 2x + 3\sqrt{x} + 4 = 0$$

$$\sqrt{x}(x+3) = -2x-4$$

$$x(x+3)^2 = (-2)^2(x+2)^2$$

$$x(x^2+6x+9) = 4(x^2+4x+4)$$

$$x^3 + 6x^2 + 9x = 4x^2 + 16x + 16$$

$$x^3 + 2x^2 - 7x - 16 = 0$$

$\therefore$ The cubic equation with roots $\alpha^2, \beta^2, \gamma^2$

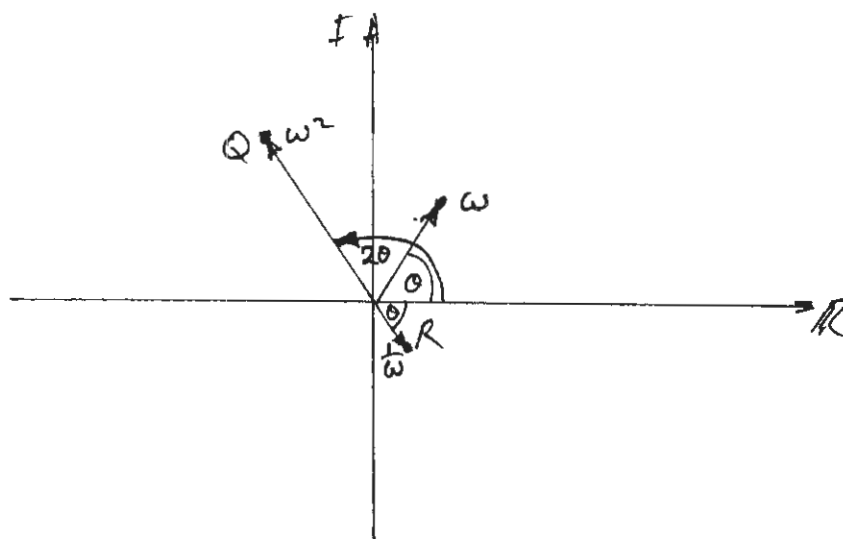
is $x^3 + 2x^2 - 7x - 16 = 0$

QUESTION 14

$$(a) \quad \omega = 2(\cos \theta + i \sin \theta) = 2 \operatorname{cis} \theta \quad \frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}$$

$$(i) \quad \begin{aligned} \omega^2 &= 4 \operatorname{cis} 2\theta \\ &= 4(\cos 2\theta + i \sin 2\theta) \end{aligned} \quad \begin{aligned} \frac{1}{\omega} &= \frac{1}{2} \operatorname{cis}(-\theta) \\ &= \frac{1}{2}(\cos(-\theta) + i \sin(-\theta)) \end{aligned}$$

(ii)



$$2\theta + \theta = \pi$$

$$\theta = \frac{\pi}{3}$$

$$\begin{aligned} \omega &= 2 \cos \frac{\pi}{3} + 2i \sin \frac{\pi}{3} \\ &= 2 \times \frac{1}{2} + i \times 2 \frac{\sqrt{3}}{2} \end{aligned}$$

$$\omega = 1 + i\sqrt{3}$$

$$= a + ib \quad \text{where } a=1, b=\sqrt{3}$$

QUESTION 14 (continued)

$$(b) \quad I_n = \int_0^a (1+x^2)^n dx \quad n=1, 2, 3, \dots$$

$$\begin{aligned} (i) \quad I_n &= \left[x(1+x^2)^n \right]_0^a - \int_0^a 2x^2 n (1+x^2)^{n-1} dx & u &= (1+x^2)^n & \frac{du}{dx} &= 2x & \frac{dy}{dx} &= 1 \\ &= (a(1+a^2)^n - 0) - 2n \int_0^a x^2 (1+x^2)^{n-1} dx & \frac{du}{dx} &= n(1+x^2)^{n-1} \times 2x & v &= x \\ &= a(1+a^2)^n - 2n \int_0^a (1+x^2 - 1)(1+x^2)^{n-1} dx & &= 2xn(1+x^2)^{n-1} \\ &= a(1+a^2)^n - 2n \int_0^a (1+x^2)^n - (1+x^2)^{n-1} dx \\ &= a(1+a^2)^n - 2n I_n + 2n I_{n-1} \end{aligned}$$

$$\therefore I_n (1+2n) = a(1+a^2)^n + 2n I_{n-1}$$

$$I_n = \frac{a(1+a^2)^n}{2n+1} + \frac{2n}{2n+1} I_{n-1}$$

$$(ii) \quad I_1 = \frac{a(1+a^2)^1}{2(1)+1} + \frac{2(1)}{2(1)+1} I_0$$

$$= \frac{a(1+a^2)}{3} + \frac{2}{3} I_0$$

$$= \frac{a+a^3}{3} + \frac{2a}{3}$$

$$= \frac{3a+a^3}{3}$$

$$I_0 = \int_0^a 1 dx$$

$$= [x]_0^a$$

$$= a$$

$$(iii) \quad \int_0^1 (1+x^2)^3 dx = I_3 \quad \text{where } a=1 \text{ and } I_n = \frac{2^n}{2n+1} + \frac{2n}{2n+1} I_{n-1}$$

$$= \frac{8}{7} + \frac{6}{7} I_2$$

$$= \frac{8}{7} + \frac{6}{7} \left(\frac{4}{5} + \frac{4}{5} I_1 \right)$$

$$= \frac{8}{7} + \frac{24}{35} + \frac{6}{7} \times \frac{4}{5} \times \frac{4}{3}$$

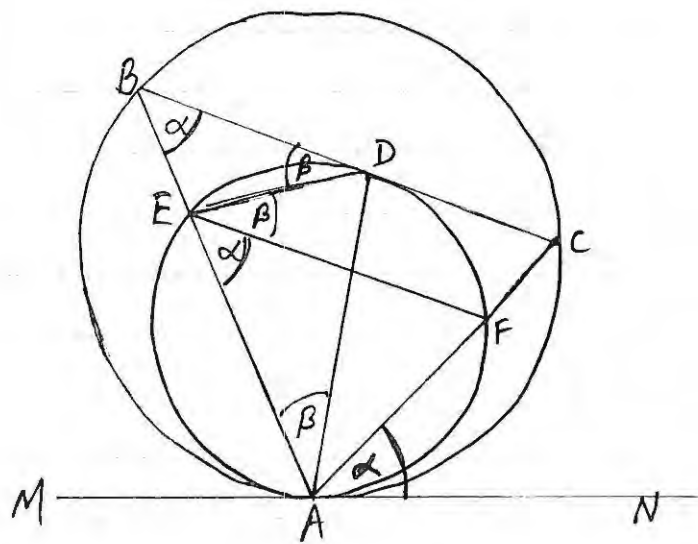
$$= \frac{8}{7} + \frac{24}{35} + \frac{32}{35}$$

$$= 2 \frac{26}{35}$$

$$\text{and } I_1 = \frac{3a+a^3}{3} = \frac{4}{3}$$

QUESTION 14 (continued)

(c)



RTP AD bisects $\angle BAC$

that is RTP $\angle BAD = \angle FAD$.

Construct EF and ED .

Let $\alpha = \angle FEA = \angle FAN$ (angle between tangent and chord equals
the angle in the alternate segment)
then $\alpha = \angle CBA = \angle CAN$ (" " " " " " " " " ")

$\therefore BC \parallel EF$ (corresponding angles $\angle CBA = \angle FEA$ are equal so parallel lines exist).

Let $\angle DEF = \beta$

Then $\angle BDE = \angle DEF = \beta$ (alternate angles in $\parallel$ lines are equal)
 $\angle BAD = \angle BDE = \beta$ (alternate segment theorem (as above))

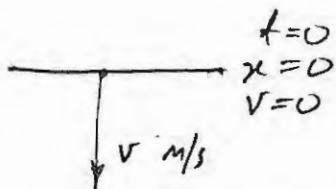
$\angle FAD = \angle DEF = \beta$ (angles standing on the same arc DF subtend equal angles at the circumference)

$$\therefore \angle BAD = \angle FAD$$

$\therefore AD$ bisects $\angle BAC$ as required.

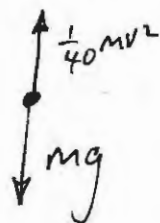
QUESTION 15

(a)



down = positive
 $g = 10 \text{ m/s}^2$
 downwards

Forces diagram



(i) Resultant Force $= m\ddot{x} = mg - \frac{1}{40}mv^2$

$$\begin{aligned}\ddot{x} &= g - \frac{1}{40}v^2 \\ &= \frac{40g}{40} - \frac{v^2}{40} \\ &= \frac{40g - v^2}{40} \\ &= \frac{400 - v^2}{40}\end{aligned}$$

$= \frac{1}{40}(400 - v^2)$ as required.

(ii) $\frac{dv}{dt} = \frac{1}{40}(400 - v^2)$

$$\frac{dt}{dv} = \frac{40}{400 - v^2}$$

$$t = 40 \int \frac{1}{400 - v^2} dv$$

$$= 40 \int \frac{1}{(20+v)(20-v)} dv$$

$$= \frac{40}{40} \int \frac{1}{20+v} + \frac{1}{20-v} dv$$

$$= \left[\ln(20+v) - \ln(20-v) \right] + c$$

$$= \ln \left(\frac{20+v}{20-v} \right) + c$$

when $t=0$, $v=0 \therefore c=0$

$$\therefore t = \ln \left(\frac{20+v}{20-v} \right)$$

$$\frac{1}{(20+v)(20-v)} = \frac{a}{20+v} + \frac{b}{20-v}$$

$$1 = a(20-v) + b(20+v)$$

$$\begin{aligned}b-a &= 0 \\ a &= b\end{aligned}$$

$$20a + 20b = 1$$

$$a+b = \frac{1}{20}$$

$$a=b = \frac{1}{40}$$

QUESTION 15 (continued)

(a) (iii) $t = \ln\left(\frac{20+v}{20-v}\right)$

$$e^t = \frac{20+v}{20-v}$$

$$(20-v)e^t = 20+v$$

$$20e^t = 20 + v(1+e^t)$$

$$20(e^t - 1) = v(1+e^t)$$

$$v = 20\left(\frac{e^t - 1}{1+e^t}\right)$$

$$= 20\left(\frac{1+e^t - 2}{1+e^t}\right)$$

$$= 20\left(1 - \frac{2}{1+e^t}\right) \quad \text{as required.}$$

(b) $P(x) = x^4 - 2x^3 - x^2 + 6x - 6$

Since $P(1-i) = 0$ $x - (1-i)$ is a factor of $P(x)$
and so is $x - (1+i)$

$$\begin{aligned} (x - (1-i))(x - (1+i)) &= (x-1+i)(x-1-i) \\ &= (x-1)^2 - i^2 \\ &= x^2 - 2x + 1 + 1 \\ &= x^2 - 2x + 2 \end{aligned}$$

$$\begin{aligned} P(x) &= (x^4 - 2x^3 - x^2 + 6x - 6) = (x^2 - 2x + 2)(x^2 + 0x - 3) \\ &= (x^2 - 2x + 2)(x^2 - 3) \\ &= (x - (1-i))(x - (1+i))(x - \sqrt{3})(x + \sqrt{3}) \end{aligned}$$

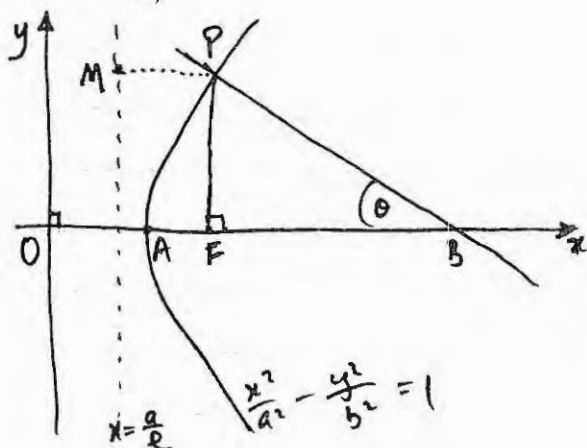
$\therefore$ the four solutions to $P(x) = 0$ are:

$$x = 1-i, 1+i, \sqrt{3}, -\sqrt{3}$$

i.e. $x = \pm\sqrt{3}, 1 \pm i$

QUESTION 15 (continued)

(c) (i)



$$A(a, 0) \quad F(ae, 0)$$

$$\begin{aligned} PF &= ePM \\ &= e \left(ae - \frac{a}{e} \right) \\ &= ae^2 - a \\ &= a(e^2 - 1) \end{aligned}$$

$$\therefore P(ae, a(e^2 - 1))$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{2x}{a^2} \div \frac{2y}{b^2} \\ &= \frac{b^2 x}{a^2 y} \end{aligned}$$

$$= (e^2 - 1) \frac{x}{y}$$

$$\text{at } P, \frac{dy}{dx} = (e^2 - 1) \frac{ae}{a(e^2 - 1)} = e$$

$$b^2 = a^2(e^2 - 1)$$

$$\frac{b^2}{a^2} = e^2 - 1$$

$\therefore$ Normal at P has gradient $= -\frac{1}{e}$

$$\therefore \tan(180 - \theta) = -\frac{1}{e}$$

$$-\tan \theta = -\frac{1}{e}$$

$$\tan \theta = \frac{1}{e} \quad \text{as required.}$$

$$\begin{aligned} \text{(ii)} \quad AF &= h = ae - a \\ h &= a(e - 1) \end{aligned}$$

$$PF = a(e^2 - 1)$$

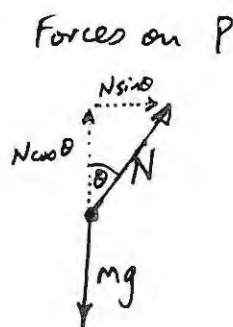
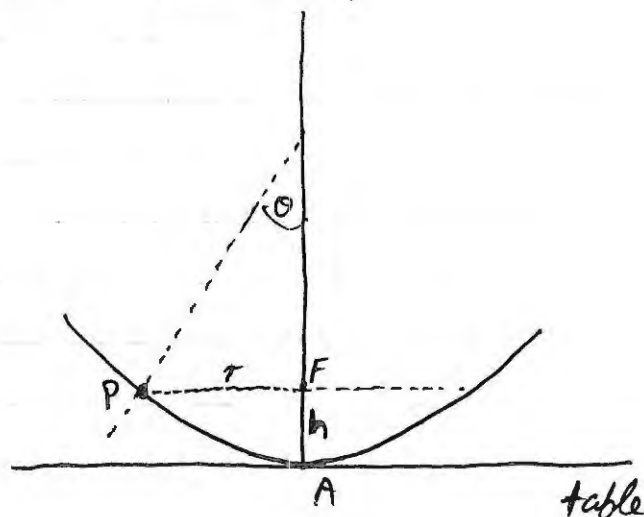
$$= a(e - 1)(e + 1)$$

$$= h(e + 1)$$

as required.

QUESTION 15 (continued)

(e) (iii)



Resolving forces Vertically

$$N \cos \theta = mg \quad \text{--- (1)}$$

Radially

$$N \sin \theta = m \omega^2 r \quad \text{--- (2)}$$

$$\textcircled{2} \div \textcircled{1} \quad \tan \theta = \frac{m \omega^2 r}{mg}$$

$$\frac{1}{e} = \frac{\omega^2}{g} h(e+1)$$

since $r = PF = h(e+1)$

$$\frac{g}{e} = \omega^2 h(e+1)$$

$$\therefore \omega^2 = \frac{g}{he(e+1)} \quad \text{as required.}$$

(iv) for a rectangular hyperbola $e = \sqrt{2}$

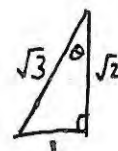
$$N \cos \theta = mg$$

$$\tan \theta = \frac{1}{e} = \frac{1}{\sqrt{2}}$$

$$N \frac{\sqrt{2}}{\sqrt{3}} = mg$$

$$N \sqrt{\frac{2}{3}} = mg$$

$$\therefore N = mg \sqrt{\frac{3}{2}}$$



QUESTION 16

(a) (i) The hyperbola $(ct, \frac{c}{t})$ intersects the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\text{when } \frac{(ct)^2}{a^2} + \frac{(\frac{c}{t})^2}{b^2} = 1$$

$$b^2 c^2 t^2 + \frac{a^2 c^2}{t^2} = a^2 b^2$$

$$b^2 c^2 t^4 + a^2 c^2 = a^2 b^2 t^2$$

$$(bc)^2 t^4 - (ab)^2 t^2 + (ac)^2 = 0$$

The number of solutions is 0, 2, or 4 = # of possible intersection points.

For the hyperbola to "TOUCH" the ellipse at P where $t=p$,

p must be a double root, and to "TOUCH" at Q

where $t=-p$, $-p$ must be a double root.

$\therefore$ the roots of the equation are $p, p, -p, -p$.

$$(ii) \sum \text{roots in pairs} = p^2 - p^2 - p^2 - p^2 - p^2 + p^2 = \frac{-(ab)^2}{(bc)^2}$$

$$-2p^2 = -\frac{a^2}{c^2}$$

$$p^2 = \frac{a^2}{2c^2}$$

$$p = \frac{a}{\sqrt{2}c}$$

$$\text{product of roots} = p^4 = \frac{(ca)^2}{(bc)^2}$$

$$p^4 = \frac{a^2}{b^2}$$

$$p^2 = \frac{a}{b}$$

$$\text{but } p^2 = \frac{a^2}{2c^2} \therefore \frac{a}{b} = \frac{a^2}{2c^2}$$

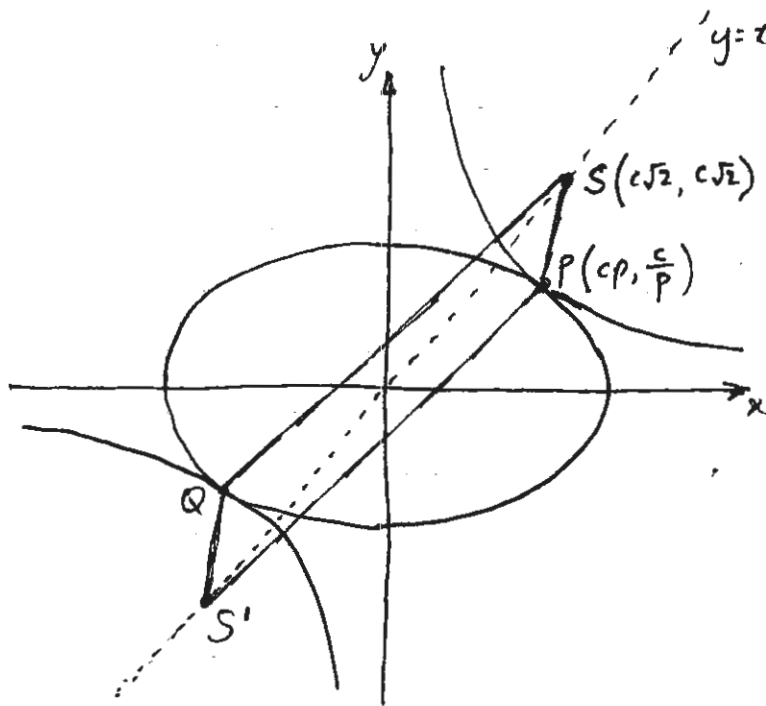
$$2ac^2 = a^2 b$$

$$\therefore ab = 2c^2$$

$$\therefore p = \frac{a}{\sqrt{2}c} \quad \text{and} \quad ab = 2c^2$$

QUESTION 16 continued

(a) (iii)



$$y=x$$

$$x-y+0=0$$

general form.

$$\text{Area of } PSQS' = 2 \times \text{Area } PSS'$$

$$= 2 \times \frac{1}{2} \times SS' \times (\perp \text{ distance from } P \text{ to } y=x)$$

$$= 2 \times \frac{1}{2} (2c + 2c) \times \left| \frac{1 \times cp - 1 \times \frac{c}{p} + 0}{\sqrt{1^2 + (-1)^2}} \right|$$

$$= 1 \times 4c \times \frac{cp - \frac{c}{p}}{\sqrt{2}}$$

$$= \frac{4c}{\sqrt{2}} \times \left(cp - \frac{c}{p} \right)$$

$$= \frac{4c}{\sqrt{2}} \times \left(\frac{a}{\sqrt{2}} - c \times \frac{c\sqrt{2}}{a} \right)$$

$$= \frac{4c}{\sqrt{2}} \times \left(\frac{a}{\sqrt{2}} - \frac{c^2\sqrt{2}}{a} \right)$$

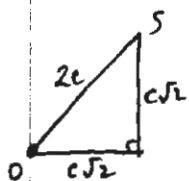
$$= 2ac - \frac{8c^3}{a}$$

$$= 2c \left(a - \frac{2c^2}{a} \right)$$

$$= 2c(a - b)$$

$$\begin{cases} p = \frac{a}{c\sqrt{2}} \\ cp = \frac{a}{\sqrt{2}} \end{cases}$$

$$\begin{cases} ab = 2c^2 \\ b = \frac{2c^2}{a} \end{cases}$$



QUESTION 16 (continued)

(b) (i) RTP $\frac{1}{a} + \frac{1}{b} \geq \frac{4}{a+b}$

i.e. RTP $\frac{1}{a} + \frac{1}{b} - \frac{4}{a+b} \geq 0$ since $P \geq Q \Leftrightarrow P - Q \geq 0$

$$\begin{aligned} \frac{1}{a} + \frac{1}{b} - \frac{4}{a+b} &= \frac{b+a}{ab} - \frac{4}{a+b} \\ &= \frac{(a+b)^2 - 4ab}{ab(a+b)} \\ &= \frac{a^2 + b^2 + 2ab - 4ab}{ab(a+b)} \\ &= \frac{a^2 + b^2 - 2ab}{ab(a+b)} \\ &= \frac{(a-b)^2}{ab(a+b)} \end{aligned}$$

≥ 0 since $(a-b)^2 \geq 0$ and $ab(a+b) \geq 0$ when $a, b > 0$.

(ii) $\frac{x+y}{2} \geq \sqrt{xy}$

$\left. \begin{matrix} x = \frac{1}{a^2} \\ y = \frac{1}{b^2} \end{matrix} \right\} \Rightarrow \frac{\frac{1}{a^2} + \frac{1}{b^2}}{2} \geq \sqrt{\frac{1}{a^2} \cdot \frac{1}{b^2}}$

$\frac{1}{a^2} + \frac{1}{b^2} \geq \frac{2}{ab}$

$\geq \frac{8}{(a+b)^2}$

Also, $\frac{a+b}{2} \geq \sqrt{ab}$

$\therefore \frac{(a+b)^2}{4} \geq ab$

$\therefore (a+b)^2 \geq 4ab$

$\therefore \frac{1}{4ab} \geq \frac{1}{(a+b)^2}$

$\therefore \frac{2}{ab} \geq \frac{8}{(a+b)^2}$

QUESTION 16 (continued)

(c) $f(x) = \frac{x^n}{e^x} \quad n > 1$

(i) $f'(x) = \frac{e^x \cdot nx^{n-1} - x^n \cdot e^x}{(e^x)^2}$

$$= \frac{e^x x^{n-1} (n - x)}{(e^x)^2}$$

$$= \frac{x^{n-1} (n - x)}{e^x}$$

(ii) stationary points when $f'(x) = 0$

$$\frac{x^{n-1} (n - x)}{e^x} = 0$$

$$x^{n-1} (n - x) = 0$$

$$x = 0 \quad \text{OR} \quad x = n$$

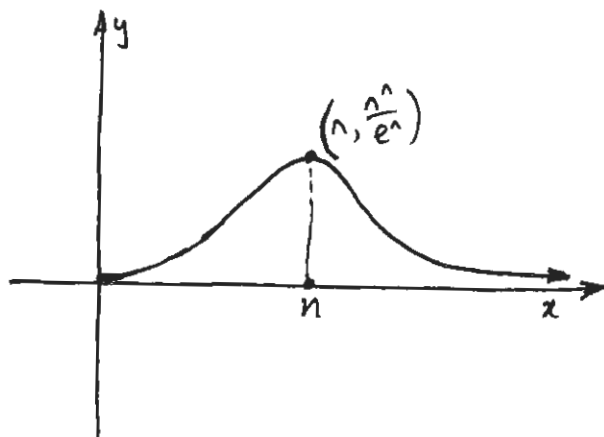
x	—	0	$\frac{n}{2}$	n	$2n$
$f'(x)$	—	0	+	0	—

gradient $\Rightarrow$ — / — \

$$\lim_{x \rightarrow \infty} \frac{x^n}{e^x} = 0^+$$

when $x = 0$ $f(x) = 0$

when $x = n$ $f(n) = \frac{n^n}{e^n}$



QUESTION 16 (continued)

(c) (iii) when $x > n$ the curve is decreasing

$\therefore$ the y-coordinate of the graph for $x > n$
is always less than the maximum value $\frac{n^n}{e^n}$

$$\therefore \frac{x^n}{e^x} < \frac{n^n}{e^n} \quad \text{for } x > n.$$

$$(iv) \quad \frac{x^n}{e^x} < \frac{n^n}{e^n} \quad \text{for } x > n$$

use $x = n+1$, $x > n$

$$\frac{(n+1)^n}{e^{n+1}} < \frac{n^n}{e^n}$$

$$\frac{(n+1)^n}{n^n} < \frac{e^{n+1}}{e^n}$$

noting $n^n > 0$
and $e^{n+1} > 0$.

$$\left(\frac{n+1}{n}\right)^n < \frac{e^n e}{e^n}$$

$$\left(\frac{n+1}{n}\right)^n < e$$

$$\therefore \left(1 + \frac{1}{n}\right)^n < e$$